46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-15** **Date: 09.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A body projected vertically upwards with a certain speed from the top of a tower reaches the ground in t_1. If it is projected vertically downwards from the same point with the same speed, it reaches the ground in t_2. Time required to reach the ground, if it is dropped from the top of the tower is

A) $\sqrt{\frac{t_1}{t_2}}$ B) $\sqrt{t_1+t_2}$ C) $\sqrt{t_1-t_2}$ D) $\sqrt{t_1t_2}$

2. When UV light of wavelength 300 *nm* is incident on the metal surface having work function 2.13*eV*, electron emission takes place. The stopping potential is (Given *hc* =

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$1240 eV\,nm$)

A) $2V$ B) $4.1V$ C) $4V$ D) $1.5V$

3. When kinetic energy of a body becomes 36 times of its original value, the percentage increase in the momentum of the body will be

A) 600 % B) 6 % C) 60 % D) 500 %

4. Pressure inside a soap bubble is greater than the pressure outside by an amount :
(Given R = Radius of bubble, S = Surface tension of bubble)

A) $\frac{S}{R}$ B) $\frac{2S}{R}$ C) $\frac{4R}{S}$ D) $\frac{4S}{R}$

5. A particle moves in a straight line so that its displacement x at any time t is given by $x^2 = 1 + t^2$. Its acceleration at any time t is x^{-n} where n = ___

A) 1 B) 2 C) 3 D) 4

6. Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth. How much it would weigh at $\frac{R}{4}$ depth under surface of earth?

A) 75 N B) 225 N C) 300 N D) 375 N

7. Given below are two statements:

Statement – I: Dimensions of specific heat is $\left[L^2T^{-2}K^{-1}\right]$.

Statement – II: Dimensions of gas constant is $\left[ML^2T^{-1}K^{-1}\right]$.

In the light of the above statements, choose the most appropriate answer from the option given below.

A) Statement – I is incorrect but Statement – II is correct

B) Both Statement – I and Statement – II are incorrect

C) Both Statement – I and Statement – II are correct

D) Statement – I is correct but Statement – II is incorrect

8. Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is

A) 12 N B) 4 N C) 1N D) 6 N

9. The number of electrons flowing per second in the filament of a 110W bulb operating at 220V is (Given e = 1.6×10^{-19} C)

A) 6.25×10^{17} B) 31.25×10^{17} C) 6.25×10^{18} D) 1.25×10^{19}

10. Energy of 10 non rigid diatomic molecules at temperature T is

A) $70K_BT$ B) $\frac{7}{2}RT$ C) $35K_BT$ D) $35RT$

11. A body of weight 200 N is suspended from a tree branch through a chain of mass 10kg. The branch pulls the chain by a force equal to (if g = 10m/s$^{B)}$

A) 200 N B) 300 N C) 100 N D) 150 N

12. The acceptor level of a p – type semiconductor is 6 eV. The maximum wavelength of light which can create a hole would be (Given hc = 1242eV nm)

A) 207 nm B) 103.5 nm C) 414 nm D) 407 nm

13. In finding out refractive index of glass slab the following observation were made through travelling microscope 50 vernier scale division = 49 MSD; 20 divisions on main scale in each cm

For mark on paper MSR = 8.45 cm, VC = 26

For mark on paper seen through slab MSR = 7.12 cm, VC = 41

For powder particle on the top surface of the glass slab MSR = 4.05 cm, VC = 1

(MSR = Main Scale Reading, VC = Vernier Coincidence)

Refractive index of the glass slab is

A) 1.24 B) 1.52 C) 1.35 D) 1.42

14. In the given electromagnetic wave $E_y = 600\sin(\omega t - kx)Vm^{-1}$, intensity of the associated light beam is in W/m^2 : (Given $\varepsilon_0 = 9\times10^{-12}C^2N^{-1}m^{-2}$)

A) 729 B) 486 C) 243 D) 972

15. In a vernier calliper, when both jaws touch each other, zero of the vernier scale shifts towards left and its 4th division coincides exactly with a certain division on main scale. If 50 vernier scale divisions equal to 49 main scale divisions and zero error in the instrument is 0.04 *mm* then how many main scale divisions are there in1*cm*?

A) 10 B) 5 C) 40 D) 20

16. A total of 48 *J* heat is given to one mole of helium kept in a cylinder. The temperature of helium increases by 2° *C*. The work done by the gas is:

Given, $R = 8.3\ J\ K^{-1}mol^{-1}$

A) 24.9 *J* B) 72.9 *J* C) 23.1 *J* D) 48 *J*

17. In a coil, the current changes from –2 *A* to +2 *A* in 0.2 *s* and induces an emf of 0.1*V*. The self inductance of the coil is:

A) 1*mH* B) 4 *mH* C) 2.5 *mH* D) 5 *mH*

18. For the thin convex lens, the radii of curvature are at 15 *cm* and 30 *cm* respectively. The focal length of the lens is 20 *cm*. The refractive index of the material is:

A) 1.2 B) 1.4 C) 1.8 D) 1.5

19. The longest wavelength associated with Paschen series is: (Given $R_H = 1.097 \times 10^7$ *SI* unit)

A) $2.973\times10^{-6}m$ B) $3.646\times10^{-6}m$ C) $1.094\times10^{-6}m$ D) $1.876\times10^{-6}m$

20. A car of 800 *kg* is taking turn on a banked road of radius 300 *m* and angle of banking 30°. If coefficient of static friction is 0.2 then the maximum speed with which car can negotiate the turn safely: $\left(g = 10m/s^2, \sqrt{3} = 1.73\right)$

A) 264 *m/s* B) 102.8 *m/s* C) 70.4 *m/s* D) 51.4 *m/s*

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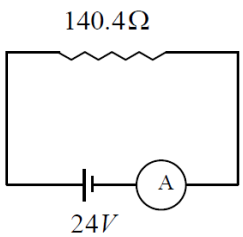
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Three balls of masses 2 *kg*, 4 *kg* and 6 *kg* respectively are arranged at centre of the edges of an equilateral triangle of side 2 *m*. The moment of intertia of the system about an axis through the centroid and perpendicular to the plane of triangle, will be _____ $kg\ m^2$

22. For a given series LCR circuit it is found that maximum current is drawn when value of variable capacitance is 2.5 *nF*. If resistance of 200Ω and 100 *mH* inductor is being used in the given circuit. The frequency of source is _____ $\times 10^3$ Hz. (*given* $\pi^2 = 10$)

23. In the given figure an ammeter *A* consists of a 240 Ω coil connected in parallel to a 10 Ω shunt. The reading of the ammeter is _____ *mA*



24. A capacitor of 10μF capacitance whose plates are separated by 10*mm* through air and each plate has area $4cm^2$ is now filled equally with two dielectric media of $K_1 = 2$, $K_2 = 3$ respectively as shown in figure. If new force between the plates is 8 *N*. The supply voltage is _____ *V*

25. Two open organ pipes of lengths 60 *cm* and 90 *cm* resonate at 6th and 5th harmonics respectively. The difference of frequencies for the given modes is _____ *Hz* (Velocity of sound in air = 333 *m/s*)

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
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26. The incorrect statements regarding enzymes are:

a) Enzymes are biocatalysts

b) Enzymes are non-specific and can catalyes different kinds of reactions.

c) Most Enzymes are globular proteins

d) Enzyme-oxidase catalyes the hydrolysis of maltose into glucose

Choose the correct answer from the option given below:

A) (b) and (c) only B) (a), (b) and (c) only

C) (b), (c) and (d) only D) (b) and (d) only

27. Match List – I with List – II

	List – I **Reaction**		**List – II** **Type of redox reaction**
a.	$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$	I)	Decomposition
b.	$2Pb(NO_3)_{2(s)} \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$	II)	Displacement
c.	$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$	III)	Disproportionation
d.	$2NO_{2(g)} + 2OH^-_{(aq)} \rightarrow NO^-_{2(aq)} + NO^-_{3(aq)} + H_2O_{(l)}$	IV)	Combination

A) $a - II, b - III, c - IV, d - I$ B) $a - IV, b - I, c - II, d - III$

C) $a - III, b - II, c - I, d - IV$ D) $a - I, b - II, c - III, d - IV$

28. Evaluate the following statements related to group 14 elements for their correctness.

a) Covalent radius decreases down the group from C to *Pb* in a regular manner

b) Electro negativity decreases from C to *Pb* down the group gradually

c) Maximum covalence of C is 4 whereas other elements can expand their covalence due to the presence of d orbitals.

d) Heavier elements do not form $\rho\pi - \rho\pi$ bonds.

e) Carbon can exhibits negative oxidation states.

Choose the correct answer from the options given below:

A) (a) and (b) only
B) (c), (d) and (e) only
C) (a), (b) and (c) only
D) (c) and (d) only

29. During the detection of acidic radical present in a salt, a student gets a pale yellow precipitate soluble with difficulty in NH_4OH solution when sodium carbonate extract was first acidified with $dil.HNO_3$ and then $AgNO_3$ solution was added. This indicates presence of:

A) CO_3^{2-}
B) I^-
C) Cl^-
D) Br^-

30. Given below are two statements:

Statement – I: PF_5 and BrF_5 both exhibit sp^3d hybridization.

Statement – II: Both SF_6 and $\left[Co\left(NH_3\right)_6\right]^{3+}$ exhibit sp^3d^2 hybridization.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement – I and Statement – II are false

B) Statement – I is true but Statement – II is false

C) Both Statement – I and Statement – II are true

D) Statement – I is false but Statement – II is true

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31. Arrange the following elements in the increasing order of number of unpaired electrons in them.

a) *Sc* b) *Cr* c) *V* d) *Ti* e) *Mn*

Choose the correct answer from the options given below:

A) $a < d < c < b < e$ B) $c < e < b < a < d$

C) $b < c < d < e < a$ D) $a < d < c < e < b$

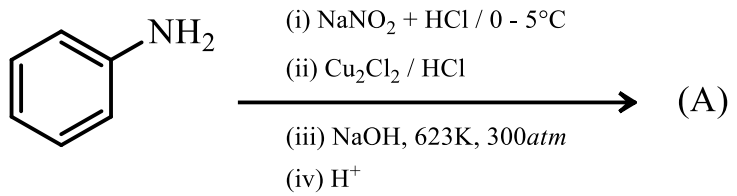
32. Consider the given reaction, identify the major product P.

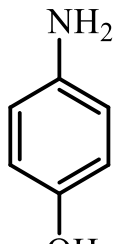
$$CH_3-COOH \xrightarrow[(iv)\ H_2O/\overline{O}H,\Delta]{(i)\ LiAlH_4\ \ (ii)\ PCC\ \ (iii)\ HCN/\overline{O}H} "P"$$

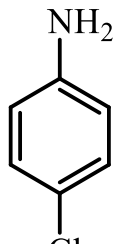
A) $CH_3-\underset{}{\overset{OH}{\overset{|}{C}H}}-COOH$ B) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-NH_2$

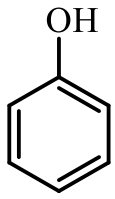
C) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2CH_3$ D) $CH_3-CH_2-CH_2-OH$

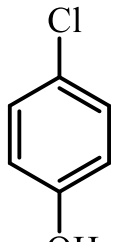
33. Identify the major product (A) in the following reaction:



A) 

B) 

C) 

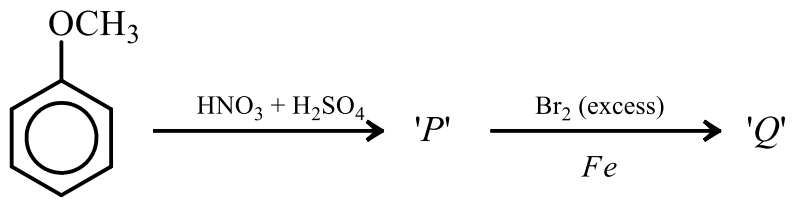
D 

34. Molality (m) of 3M aqueous solution of *NaCl* is:

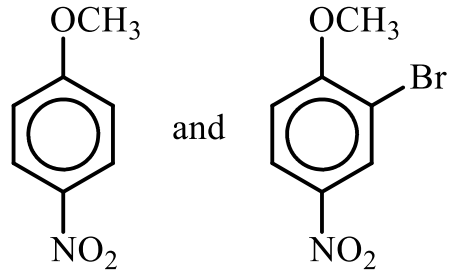
(Given: Density of solution = 1.25 gmL^{-1}, Molar mass in $g\ mol^{-1}$: *Na* – 23, *Cl* – 35.5)

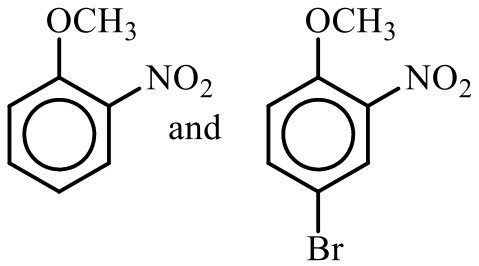
A) 3.85m B) 2.79m C) 1.90m D) 2.90m

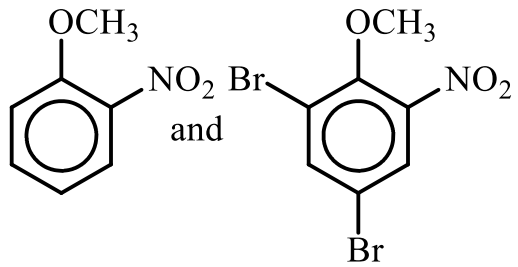
35. The major products formed:

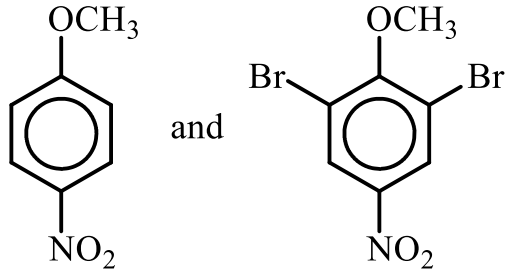


P and *Q* respectively are:

A) 

B) 

C) 

D) 

36. The ratio $\frac{K_P}{K_C}$ for the reaction: $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons CO_{2(g)}$ is:

A) $(RT)^{1/2}$ B) RT C) 1 D) $\frac{1}{\sqrt{RT}}$

37. The correct statement among the following, for a "chromatography" purification method is:

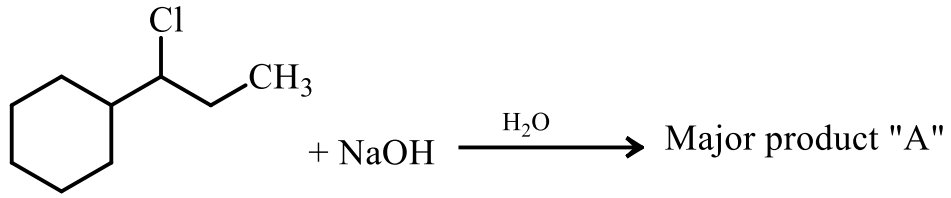
A) R_f is an integral value

B) Organic compounds run faster than solvent in the thin layer chromatographic plate

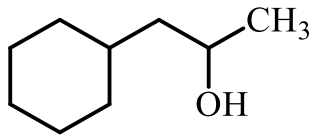
C) R_f of a polar compound is smaller than that of a non-polar compound.

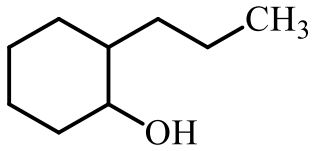
D) Non-polar compounds are retained at top and polar compounds come down in column chromatography.

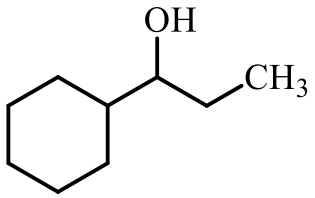
38.

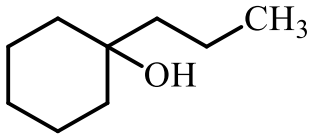


Consider the above chemical reaction. product “A” is:

A) 

B) 

C) 

D) 

39. How can an electrochemical cell be converted into an electrolytic cell?

A) Applying an external opposite potential lower than E^0_{cell}.

B) Reversing the flow of ions in salt bridge

C) Exchanging the electrodes at anode and cathode

D) Applying an external opposite potential greater than E^0_{cell}.

40. Match List – I with List – II.

	List-I **Tetrahedral Complex**		**List – II** **Electronic configuration**
a.	$TiCl_4$	I)	e_g^2, t_{2g}^0
b.	$[FeO_4]^{2-}$	II)	e_g^4, t_{2g}^3
c.	$[FeCl_4]^-$	III)	e_g^0, t_{2g}^0
d.	$[CoCl_4]^{2-}$	IV)	e_g^2, t_{2g}^3

Choose the correct answer from the options given below:

A) $a-III, b-IV, c-II, d-I$ B) $a-III, b-I, c-IV, d-II$

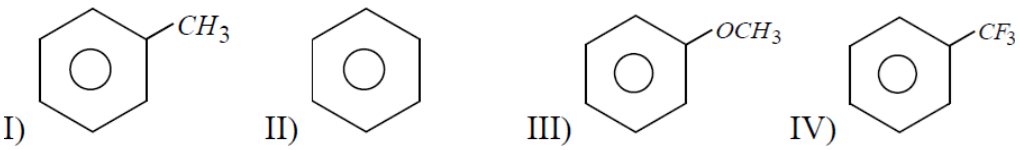
C) $a-I, b-III, c-IV, d-II$ D) $a-IV, b-III, c-I, d-II$

41. The number of ions from the following that are expected to behave as oxidizing agent is:

$Sn^{4+}, Sn^{2+}, Pb^{2+}, Tl^{3+}, Pb^{4+}, Tl^{+}$

A) 4 B) 3 C) 1 D) 2

42.



The correct arrangement for decreasing order of electrophilic substitution for above compounds is:

A) II > IV > III > I B) III > IV > II > I

C) IV > I > II > III D) III > I > II > IV

43. Match List – I with List – II.

	List – I **Alkali Metal**		**List – II** **Emission Wavelength in** *nm*
a.	*Li*	I)	589.2
b.	*Na*	II)	455.5
c.	*Rb*	III)	670.8
d.	*Cs*	IV)	780.0

Choose the correct answer from the options given below:

A) $a-I, b-IV, c-III, d-I$ B) $a-II, b-IV, c-III, d-I$

C) $a-III, b-I, c-IV, d-II$ D) $a-IV, b-II, c-I, d-III$

44. The correct IUPAC name of $\left[PtBr_2\left(PMe_3\right)_2\right]$ is:

A) bis [bromotrimethylphosphine] platinum (II)

B) dibromodi (trimethylphosphine) platinum (II)

C) bis (trimethylphosphine) dibromoplatinum (II)

D) dibromobis(trimethylphosphine) platinum (II)

45. The incorrect statement regarding the geometrical isomers of 2-butene is:

A) cis-2-butene and trans-2-butene are not inter convertible at room temperature.

B) cis-2-butene has less dipole moment than trans-2-butene

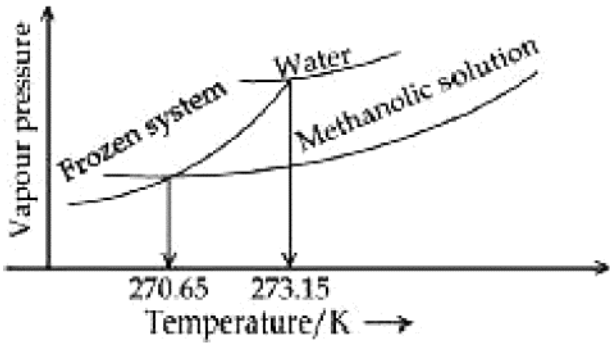
C) cis-2-butene and trans-2-butene are stereo isomers.

D) trans-2-butene is more stable than cis-2-butene

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. When 'x'$\times 10^{-2}$ mL methanol (molar mass = 32g; density = 0.792 $g/cm^{C)}$ is added to 100 mLwater (density = 1$g/cm^{C)}$, the following diagram is obtained.



[Given: Molal freezing point depression constant of water at 273.15 K is 1.86 $K\ kg\ mol^{-1}$]. Find the value of 'x'?

47. Consider the following reactions

$$NiS + HNO_3 + HCl \rightarrow A + NO + S + H_2O$$

$$A + NH_4OH + \begin{array}{c} H_3C - C = N - OH \\ | \\ H_3C - C = N - OH \end{array} \longrightarrow B + NH_4Cl + H_2O$$

The number of protons that do not involve in hydrogen bonding in the product B is ____.

48. Among VO_2^+, MnO_4^- and $Cr_2O_7^{2-}$, the spin-only magnetic moment value of the species with least oxidizing ability is ____ BM (nearest integer).

(Given atomic number V = 23, Mn = 25, Cr = 2D)

49. An amine (X) is prepared by ammonolysis of benzyl chloride. On adding p-toluenesulphonyl chloride to it the solution remains clear. Molar mass of the amine (X) formed is ____ $g\ mol^{-1}$.

(Given molar mass in $g\ mol^{-1}$ C :12, H :1, O:16, N :1D)

50. Consider the two different first order reactions given below

$A + B \rightarrow C$ (*Reaction* A)

$P \rightarrow Q$ (*Reaction* B)

The ratio of the half life of Reaction 1: Reaction 2 is 5: 2. If t_1 and t_2 represent the time taken to complete 2/3rd and 4/5th of Reaction 1 and Reaction 2, respectively, then the value of the ratio t_1:t_2 is ____$\times 10^{-1}$ (nearest integer).

[*Given*: $\log_{10}$ (C) = 0.477 *and* $\log_{10}$ (5) = 0.699]

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $P(\alpha,\beta,\gamma)$ be the image of the point $Q(3, -3, A)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point (2, 5, –A). If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to:

A) 81 B) 36 C) 18 D) 72

52. A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to:

A) 150 B) 160 C) 180 D) 125

53. Suppose the solution of the differential equation $\frac{dy}{dx}=\frac{(2+\alpha)x-\beta y+2}{\beta x-2\alpha y-(\beta\gamma-4\alpha)}$ represents a circle passing through the origin. Then the radius of this circle is :

A) $\sqrt{17}$ B) 2 C) $\frac{1}{2}$ D) $\frac{\sqrt{17}}{2}$

54. Let α,β be roots of $x^2+\sqrt{2}x-8=0$. If $U_n=\alpha^n+\beta^n$, then $\frac{U_{10}+\sqrt{2}U_9}{2U_8}$ is equal to__

A) 1 B) 4 C) 6 D) 7

55. $\lim_{n\to\infty}\frac{(1^2-1)(n-1)+(2^2-2)(n-2)+....+((n-1)^2-(n-1)).1}{(1^3+2^3+...+n^3)-(1^2+2^2+...+n^2)}$

A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{1}{3}$

56. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}|\geq 6, \vec{a}.\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to:

A) $\frac{3}{2}\sqrt{6}$ B) $\frac{9}{2}\left(6+\sqrt{6}\right)$ C) $\frac{9}{2}\left(6-\sqrt{6}\right)$ D) $\frac{3}{2}\sqrt{3}$

57. If z_1, z_2 are two distinct complex number such that $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\overline{z}_2}\right|=2$, then

A) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1

C) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius1

D) both z_1 and z_2 lie on the same circle.

58. If all the words with or without meaning made using all the letters of the word "NAGPUR: are arranged as in a dictionary, then the word at 315th position in this arrangement is:

A) NRAPUG B) NRAGUP C) NRAGPU D) NRAPGU

59. Let $f\left(x\right)=\frac{1}{7-\sin 5x}$ be a function defined on R. Then the range of the function $f(x)$ is equal to:

A) $\left[\frac{1}{8},\frac{1}{5}\right]$ B) $\left[\frac{1}{7},\frac{1}{6}\right]$ C) $\left[\frac{1}{7},\frac{1}{5}\right]$ D) $\left[\frac{1}{8},\frac{1}{6}\right]$

60. If the function $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $x=\frac{1}{e}$ then:

A) $(2e)^{\pi}>\pi^{(2e)}$ B) $e^{\pi}<\pi^{e}$ C) $e^{\pi}>\pi^{e}$ D) $e^{2\pi}<(2\pi)^{e}$

61. Let $A = \{1,2,3,4,5\}$, Let R be a relation on A defined by xRy if and only if $4x \le 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A \times A$ that are required to be added to R to make it a symmetric relation. Then $m + n$ is equal to:

A) 24 B) 25 C) 26 D) 23

62. Let [t] denote the largest integer less than or equal to t.

If $\int_0^3\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx=a+b\sqrt{2}-\sqrt{3}-\sqrt{5}+c\sqrt{6}-\sqrt{7}$, where $a, b, c \in$ Z then a + b + c is equal to ___

A) 23 B) 21 C) 25 D) 26

63. If A is a square matrix of order 3 such that det $(A) = 3$ and

$\det\left(adj\left(-4adj\left(-3adj\left(3adj(2A)^{-1}\right)\right)\right)\right)=2^m3^n$, then $m + 2n$ is equal to:

A) 2 B) 4 C) 6 D) 3

64. Let $0\le r\le n$. If ${}^{n+1}C_{r+1}:{}^{n}C_{r}:{}^{n-1}C_{r-1}=55:35:21$ then $2n + 5r$ is equal to:

A) 50 B) 62 C) 55 D) 60

65. If $\int\frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx=\frac{1}{12}\tan^{-1}(3\tan x)+$ constant, then the maximum value of $a \sin x + b \cos x$, is

A) $\sqrt{42}$ B) $\sqrt{41}$ C) $\sqrt{39}$ D) $\sqrt{40}$

66. If the locus of the point, whose distance from the point (2,A) and (1,C) are in the ratio 5:4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to:

A) 37 B) 437 C) 5 D) -27

67. Suppose for a differentiable function $h, h(0)=0, h(1)=1$ and $h'(0)=h'(1)=2$.

If $g(x)=h(e^x)e^{h(x)}$, then $g'(0)$ is equal to :

A) 8 B) 5 C) 3 D) 4

68. If the area of the region $\left\{(x,y), \frac{a}{x^2} \le y \le \frac{1}{x}, 1 \le x \le 2, 0 < a < 1\right\}$ is $(\log_e 2) - \frac{1}{7}$ then the value of 7a – 3 is equal to:

A) 0 B) – 1 C) 1 D) 2

69. Let ABC be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle ABC and the same process is repeated infinitely many times. If P is the sum of perimeters and Q is be the sum of areas of all the triangles formed in this process, then:

A) $P^2 = 36\sqrt{3}Q$ B) $P^2 = 72\sqrt{3}Q$ C) $P^2 = 6\sqrt{3}Q$ D) $P = 36\sqrt{3}Q^2$

70. If three letters can be posted to any one of the 5 different addresses, then the probability that the three letters are posted to exactly two addresses is:

A) $\frac{12}{25}$ B) $\frac{6}{25}$ C) $\frac{4}{25}$ D) $\frac{18}{25}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. If the shortest between the lines $\frac{x-\lambda}{3} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $\frac{x+2}{-3} = \frac{y+5}{2} = \frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$ then the largest possible value of $|\lambda|$ is equal to ____

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72. The length of the latus rectum and directrices of a hyperbola with eccentricity e are 9 and $x = \pm\frac{4}{\sqrt{3}}$, respectively. Let the line $y - \sqrt{3}x + \sqrt{3} = 0$ touch this hyperbola at (x_0, y_0). If m is the product of the focal distances of the point (x_0, y_0) then $4e^2 + m$ is equal to ____

73. From a lot of 12 items containing 3 defective, a sample of 5 items is drawn at random. Let the random variable X denote the number or defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where gcd(m, n) = 1, then $n - m$ is equal to ___

74. If the system of equations

$2x + 7y + \lambda z = 3$

$3x + 2y + 5z = 4$

$x + \mu y + 32z = -1$

has infinitely many solutions, then $(\lambda - \mu)$ is equal to ____

75. Let [t] denote the greatest integer less than or equal to t. Let $f : [0, \infty) \to R$ be a function defined by $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$, Let S be the set of all points in the interval $[0, 8]$ at which f is not continuous. Then $\sum_{a \in S} a$ is equal to ____